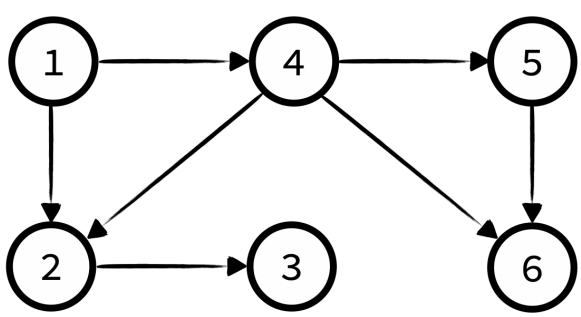
Princess Sumaya University for Technology - King Hussein School for Computing Sciences
CS11313 - **Algorithms Design and Analysis** - **Fall** 2021

Quiz # 3

Name:	**ID:**

A. Mark each of the following with `True` **or** `False`**.** `(2 points)`

- [ ] A simple directed graph with $V(V-1)$ edges is a *strongly connected* graph.
- [ ] An adjacency list is more space-efficient for dense graphs than an adjacency matrix.
- [ ] Checking if two vertices are adjacent in a *complete* simple graph represented using an adjacency list requires $\Theta(E)$ operations.
- [ ] Transposing a graph represented as an adjacency matrix requires $\Theta(V^2)$ operations.

B. Answer the following questions using the graph shown below.

- Draw a DFS tree that can be produced if a DFS is run from vertex 1. Mark the discovery and finish time on each vertex in the graph. `(3 points)`
- How many strongly connected components does the graph have? `(1 points)`
- List the vertices in topological order. `(2 points)`